

Unit 7, Lesson 7: Factoring Perfect Square Trinomials



Benchmark

MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real number system.



Learning Target

☐ I can factor perfect square trinomials.



7.7.1 Warm-Up: Critical Thinking

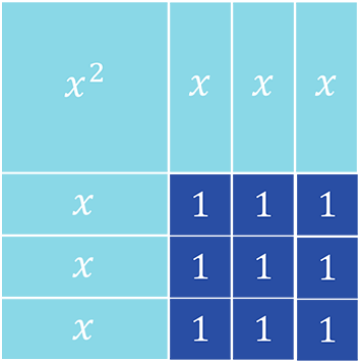
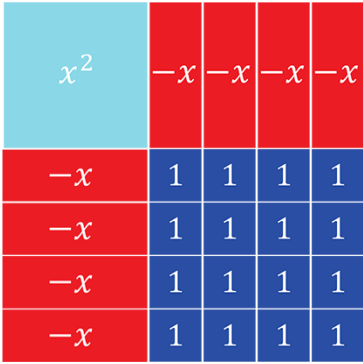
1. Fill in the blanks with an integer so that the quadratic expression is factorable. Then, complete the factored form.

Quadratic Expression	Factored Form
$x^2 + \underline{\hspace{1cm}}x + 36$	$(x + \underline{\hspace{1cm}})(x + \underline{\hspace{1cm}})$
$x^2 - \underline{\hspace{1cm}}x - 36$	$(x - \underline{\hspace{1cm}})(x + \underline{\hspace{1cm}})$
$x^2 + \underline{\hspace{1cm}}x - 36$	$(x - \underline{\hspace{1cm}})(x + \underline{\hspace{1cm}})$



7.7.2 Collaboration: Perfect Square Trinomials

1. The quadratic expressions represented by algebra tiles are perfect square trinomials.

A	B
	

- a. Discuss with your partner why these quadratic expressions are called perfect square trinomials.

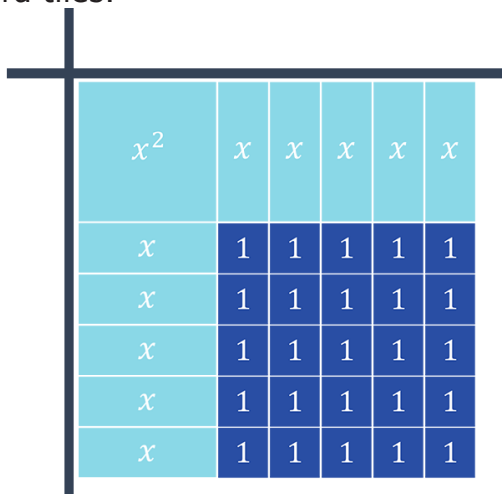
- b. Write the trinomials represented by the algebra tiles.

Trinomial A	Trinomial B

- c. Use the algebra tile representations to write the factored form of each trinomial.

Factored Form of Trinomial A	Factored Form of Trinomial B

2. The quadratic expression $x^2 + 10x + 25$ is represented by the algebra tiles.



- a. Pranjal used the algebra tiles to write the factored form of $x^2 + 10x + 25$ as $(x + 5)^2$.

Check Pranjal's work by completing the steps.

$$(x + 5)^2 = (x + 5)(x + 5)$$

$$(x + 5)(x + 5) = \underline{\hspace{2cm}}$$

- b. Discuss with your partner how writing the factored form of the quadratic expression $x^2 + 10x + 25$ as $(x + 5)^2$ relates to the expression being called a perfect square trinomial. Summarize your discussion.

3. Work with your partner to determine which quadratic expressions are perfect square trinomials.

Expression	Algebra Tiles
$4x^2 - 9$ Is it a perfect square trinomial? ☐ Yes ☐ No	
$4x^2 + 12x + 9$ Is it a perfect square trinomial? ☐ Yes ☐ No	

$4x^2 - 12x + 9$ Is it a perfect square trinomial? Yes No	<table><tr><td>x^2</td><td>x^2</td><td>$-x$</td><td>$-x$</td><td>$-x$</td></tr><tr><td>x^2</td><td>x^2</td><td>$-x$</td><td>$-x$</td><td>$-x$</td></tr><tr><td>$-x$</td><td>$-x$</td><td>1</td><td>1</td><td>1</td></tr><tr><td>$-x$</td><td>$-x$</td><td>1</td><td>1</td><td>1</td></tr><tr><td>$-x$</td><td>$-x$</td><td>1</td><td>1</td><td>1</td></tr></table>	x^2	x^2	$-x$	$-x$	$-x$	x^2	x^2	$-x$	$-x$	$-x$	$-x$	$-x$	1	1	1	$-x$	$-x$	1	1	1	$-x$	$-x$	1	1	1
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x^2	x^2	$-x$	$-x$	$-x$																						
$-x$	$-x$	1	1	1																						
$-x$	$-x$	1	1	1																						
$-x$	$-x$	1	1	1																						
$9x^2 + 6x + 1$ Is it a perfect square trinomial? Yes No	<table><tr><td>x^2</td><td>x^2</td><td>x^2</td><td>x</td></tr><tr><td>x^2</td><td>x^2</td><td>x^2</td><td>x</td></tr><tr><td>x^2</td><td>x^2</td><td>x^2</td><td>x</td></tr><tr><td>x</td><td>x</td><td>x</td><td>1</td></tr></table>	x^2	x^2	x^2	x	x^2	x^2	x^2	x	x^2	x^2	x^2	x	x	x	x	1									
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x^2	x^2	x^2	x																							
x^2	x^2	x^2	x																							
x	x	x	1																							
$9x^2 - 1$ Is it a perfect square trinomial? Yes No	<table><tr><td>x^2</td><td>x^2</td><td>x^2</td><td>x</td></tr><tr><td>x^2</td><td>x^2</td><td>x^2</td><td>x</td></tr><tr><td>x^2</td><td>x^2</td><td>x^2</td><td>x</td></tr><tr><td>$-x$</td><td>$-x$</td><td>$-x$</td><td>-1</td></tr></table>	x^2	x^2	x^2	x	x^2	x^2	x^2	x	x^2	x^2	x^2	x	$-x$	$-x$	$-x$	-1									
x^2	x^2	x^2	x																							
x^2	x^2	x^2	x																							
x^2	x^2	x^2	x																							
$-x$	$-x$	$-x$	-1																							

4. Describe some key characteristics of perfect square trinomials by relating both the representation of a quadratic expression using algebra tiles and the trinomial.



7.7.3 Guided Instruction: Factoring Perfect Square Trinomials

A **perfect square trinomial** is a trinomial of the form $a^2 + 2ab + b^2$ or $a^2 - 2ab + b^2$.

1. The quadratic expression $4x^2 + 12x + 9$ is a perfect square trinomial. Rewrite the trinomial in the form $a^2 + 2ab + b^2$.

Perfect square trinomials of the form $a^2 + 2ab + b^2$ can be factored as $(a + b)^2$.

2. Factor $4x^2 + 12x + 9$.

3. The quadratic expression $9x^2 - 12x + 4$ is a perfect square trinomial. Rewrite the trinomial in the form $a^2 - 2ab + b^2$.

Perfect square trinomials of the form $a^2 - 2ab + b^2$ can be factored as $(a - b)^2$.

4. Factor $9x^2 - 12x + 4$.





7.7.4 Guided Practice: Factoring Perfect Square Trinomials

- Place a check mark in the appropriate columns of the table to indicate whether or not each quadratic expression is a perfect square trinomial. If it is a perfect square, write its factored form as $(a + b)^2$ or $(a - b)^2$.

Quadratic Expression	First Term is a Perfect Square	Third Term is a Perfect Square	Second Term is $2ab$ or $-2ab$	Factored Form
$2g^2 + 6g + 9$				
$16y^2 - 56y + 49$				
$b^2 - 26b + 25$				
$4w^2 - 16w + 8$				
$9x^2 - 4$				
$q^2 - 2qz + z^2$				



7.7.5 Your Turn!

- Factor $16h^2 + 88h + 121$.

- Factor $k^2 - 10k + 25$.



7.7.6 Wrap-Up: Reflection

1. Choose the emoji that reflects your understanding of factoring perfect square trinomials.



0 1 2 3 4 5 6 7 8 9 10

2. Complete the statement.

I chose this emoji because _____



Unit 7, Lesson 8: Factoring the Difference of Two Squares



Benchmark

MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real number system.



Learning Targets

- ☐ I can factor a difference of two squares.
- ☐ I can rewrite a polynomial expression as a product of polynomials by factoring.